

Auteur: Thijs Van Schel
 Studierichting: Informatica
 Oefeningenreeks 5A nr. 8.2

$$\int_{-\frac{1}{2}}^1 \sqrt{1-x^2} dx$$

$$x = \sin t \Rightarrow t = \arcsin x \Rightarrow dx = \cos t dt$$

O.I.:

$$\begin{aligned} & \int \cos t \sqrt{1 - \sin^2 t} dt \\ &= \int \cos t \sqrt{\cos^2 t} dt \\ &= \int \cos^2 t dt \\ &= \frac{1}{2} \int 1 + \cos(2t) dt \\ &= \frac{1}{2} \int t^0 dt + \frac{1}{2} \int \cos(2t) dt \\ &= \frac{t}{2} + \frac{1}{4} \sin(2t) + C \end{aligned}$$

B.I.:

$$\begin{aligned} x = 1 &\Rightarrow t = \arcsin 1 = \frac{\pi}{2} \\ x = -\frac{1}{2} &\Rightarrow t = -\frac{\pi}{6} \\ \left[\frac{t}{2} + \frac{1}{4} \sin(2t) \right]_{-\frac{\pi}{6}}^{\frac{\pi}{2}} \\ &= \frac{\pi}{4} - \left(-\frac{\pi}{12} - \frac{\sqrt{3}}{8} \right) = \frac{3\pi}{12} + \frac{\pi}{12} + \frac{\sqrt{3}}{8} = \frac{\pi}{3} + \frac{\sqrt{3}}{8} \end{aligned}$$

References